

Douglas Zhou

Department of Psychology and Behavioral Sciences, Zhejiang University

Email: touchou@zju.edu.cn | Phone: +86 188-5812-0289 | Homepage: <https://douglaszhou28.github.io>

EDUCATION

Undergraduate: Zhejiang University (QS global rank 47th), Hangzhou, China

09/2023 – 06/2027

- Overall GPA: 4.17/4.3

Major: Bachelor of Science in Psychology (Honor's Program)

- **Courses:** Data Science for Psychology, Techniques in Psychological Research, Signal and Cognitive System, Psycholinguistics, Experimental Psychology, Psychometrics, etc.
- **Additional Courses:** Introduction to Natural Language Processing, Contemporary Linguistics, Machine Learning (Coursera) etc.

Minor: Mathematics and Applied Mathematics

- **Courses:** Calculus (Honor), Linear Algebra (Honor), Probability Theory (Honor), Mathematical Statistics, etc.

Exchange: University of California, Berkeley, Berkeley, USA

Summer 2025

- **Courses:** Introduction to Cognitive Science (A+), Introduction to How the Brain Works (A)

RESEARCH INTERESTS

Cognitive neuroscience of language; computational models of language processing; language and thought; human-LLMs comparison in language cognition; philosophy of language.

RESEARCH EXPERIENCE

Project : Understanding Heterogeneity in Neural Activity of Sentence Processing Using Word Embeddings

10/2024 – Present

(Leader & Core Collaborator | Advisor: Prof. Xiang-Zhen Kong | Cognomics Lab @ZJU)

- Investigated how the brain integrates multi-dimensional cognitive features during silent reading by collecting task-fMRI and 10 psycholinguistic ratings across 1,000 semantically diverse sentences. Rapid, single-trial beta estimations were optimized via **GLMsingle**.
- Built ROI-wise **ridge regression encoding models** for 384 ROIs across the whole brain (defined by AICHA atlas) using Sentence-Transformer based LLM embeddings and successfully predicted neural responses to sentences by activations across the core language network (temporal lobe, IFG), PFC and the ventral visual pathway (visual cortex, FFA).
- Further evaluated 11 distinct hypothesis-driven Model RDMs (1 LLM embedding + 10 psycholinguistic features) against Neural RDMs across extracted ROIs to see which regions respond to which features.
- Mapped specific brain regions to distinct linguistic profiles, providing neurocomputational evidence for the functional heterogeneity of sentence comprehension.

Project: An MEG Investigation of Perceptual and Semantic Constraints on Language Processing

07/2026 – Present

(Leader & Core Collaborator | Advisor: Prof. Liina Pykkänen | Neurolinguistics Lab @NYU)

- Investigating whether limits of human sentence comprehension arise from **perceptual constraints** on visual input length or **semantic constraints** on information density.
- Designing an MEG experiment using **rapid parallel visual presentation (RPVP)**, in which natural stories are presented in chunks with manipulated chunk length and information density through different font conditions.
- Implementing three stimulus conditions -- Coherent Story, ChunkList (chunks shuffled within a story), and WordList (words shuffled within a story) -- to examine the neural responses associated with discourse coherence and syntactic structure.
- Localizing brain regions and temporal windows associated with **structure effects** (Story vs. ChunkList: coherence; ChunkList vs. WordList: syntax) and increasing presentation speed to distinguish whether comprehension limits arise from physical length or information density.

Project: The Mental Number Line is Logarithmic Rather than Linear in the Human Intraparietal Sulcus

04/2026 – 07/2026

(Independent Researcher | Self-initiated and self-directed)

- Examined the **representational geometry** of numerical magnitude in the human intraparietal sulcus (IPS) to **test competing linear and logarithmic models of the mental number line**.
- Quantified the stability of neural representational geometry across individuals by integrating **Principal Coordinate Analysis (PCoA)** with **Procrustes alignment** to evaluate cross-subject convergence.
- Constructed subject-specific IPS regions of interest and generated neural representational dissimilarity matrices (RDMs) using **ROI-based RSA** on fMRI data, demonstrating that neural representations were better explained by logarithmic than linear number-line models.
- Provided evidence that number representation is stable across individuals, and numeric magnitude is encoded on a logarithmically rather than linearly compressed scale in the human IPS.

Project: *Source Languages Systematically Shape the Lexico-Grammatical Profiles of Translated English*
(Independent Researcher | Self-initiated and self-directed)

04/2026 – 07/2026

- Explored how the translated language is shaped by the negotiation between source languages and target languages.
- Extracted 67 **Biber lexico-grammatical features** for each of the 16 translated English varieties from the OPUS-100 corpus and related cross-linguistic variation to WALS typological distances to test whether a source language's distance from English makes its translated version closer to or farther from the English baseline.
- The results showed that source languages systematically shape translated texts through two different mechanisms: **Direct Structural Mapping** (where close languages trigger surface-level interference) and **Cognate Camouflage** (where distant languages undergo deep reconstruction).

Project: *Optimizing Decay and Rehearsal in a Computational Model of the Phonological Loop*
(Leader & Core Collaborator | Advisor: Dr. Christopher Kymn @UCB)

07/2025 – 08/2025

- Designed and optimized a **computational model** of the phonological loop in working memory framework, improving decay and rehearsal mechanisms to better capture memory retention processes.
- Replaced the original linear decay function with an exponential decay function; incorporated **position-dependent encoding strength** and **associative retrieval (measured by cosine similarity between word vectors from GloVe)** mechanisms.
- Developed a systematic search strategy to define model parameters (e.g., decay rate, encoding strength) and evaluated model fit using root mean squared deviation (RMSD) between empirical data and model predictions.
- Finally, the results demonstrated that these modifications significantly improved model performance, leading to better alignment with experimental data.

MANUSCRIPTS

- **Zhou, D.**, Tao X., & Kong, X. (in progress). *CSCSV-fMRI: A Semantically Diverse Mandarin Sentence-fMRI Dataset with Multidimensional Human Ratings*.
- Han, D., **Zhou, D.**, Herzberg, O., Ossmy, O., Tamis-LeMonda, C., & Adolph, K. E. (in progress). *Estimating the Rate of Behavior: Optimal Combination of Sample Size and Observation Duration*.

CONFERENCE PRESENTATIONS

- **Zhou, D.**, Tao X., Duan Y., Zhang F., Guo, D., & Kong, X. (2026, June 14). *Embedding-based Prediction of Brain Responses Reveals Emotional Effects in Sentence Processing* [Poster Presentation]. **Organization for Human Brain Mapping Annual Meeting**, Bordeaux, France.
- **Zhou, D.** (2026, April 26). *‘Can’ Psychology Be a Science? Reflections on Boundaries and Methodology*. 2050@2026 Global Youth Gathering, Yunqi Town, Hangzhou, China.
- **Zhou, D.**, Zhang F., Guo, D., & Kong, X. (2025, September 20). *Understanding Heterogeneity in Neural Activity of Sentence Processing Using Word Embeddings* [Oral Presentation]. **Undergraduate Research Symposium**, Zhejiang University, Hangzhou, China.
- **Zhou, D.**, Guo, D., & Kong, X. (2025, June 1). *CSCSV: Chinese Sentence Corpus with Semantic Variety* [Poster Presentation]. **International Workshop on Cross-linguistic Databases and Norms**, East China Normal University & The Hong Kong Polytechnic University, Shanghai, China.

PATENT

- Kong, X., **Zhou, D.**, Li, N., & Pu, Y. (2025). *A Method for Selecting Language Function Rehabilitation Corpus Based on Semantic Embedding and Neural Encoding*. Chinese Patent Application No.CN 202511371878.0, filed September 24, 2025.

GRANTS

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|---|------|
| • Undergraduate Future Academic Rising Star Program, Zhejiang University (¥20,000) | 2026 |
| • National Innovation Project, Chu Kochen Honors College Research Training Program, Zhejiang University (¥15,000) | 2025 |

HONORS & AWARDS

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| • Student Volunteer Award, Zhejiang University | 2026 |
| • Second Prize Scholarship of Zhejiang University | 2025 |
| • Third Prize Scholarship of Zhejiang University | 2024 |
| • First Prize, National High School Biology Olympiad (Zhejiang Division) | 2021, 2022 |

SKILLS

- **Experimental Techniques:** MRI, MEG, EEG, eye-tracking, VR, behavioral experiments
- **Software Techniques:** Python, MATLAB, SPM, FreeSurfer, MNE-python, Pytorch, PsychoPy, R, Praat, Linux, Coding Agent
- **Language:** English (TOEFL: 107), Mandarin (native)
- **Hobbies:** Western philosophy (especially Wittgenstein), film, literature, music, Chinese calligraphy